

GCSE Biology Paper 1 Guide: Made for ESL Students



Welcome to Your Paper 1 ESL Guide!

🌟 Welcome to Your Organisation ESL Guide!

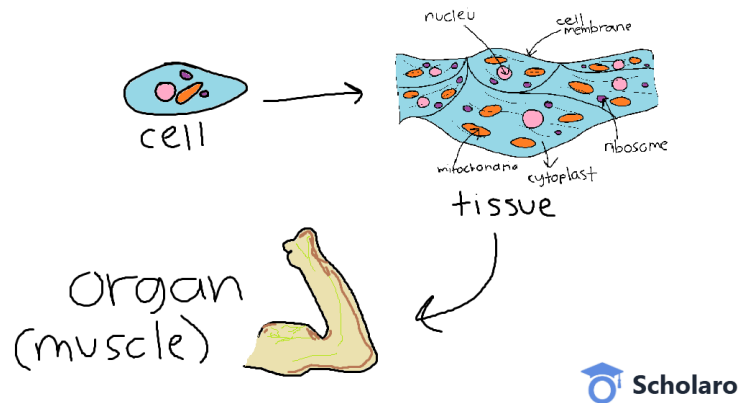
🎯 Purpose of This Guide

This guide helps ESL students understand the second topic in GCSE AQA Biology: **Organisation**. You'll learn how cells build the body, how organs work together, how food is digested, how the heart pumps blood, and how plants transport water and food.

📖 What You Will Learn

- 🧩 Levels of organisation (cells → tissues → organs → systems)
 - 🍴 Human digestion and enzymes
 - ❤️ The heart and circulatory system
 - 🏆 Blood and its parts
 - 🚑 Disease, health, and lifestyle effects
 - 🌿 Plant tissues and transport
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🧩 1. Levels of Organisation



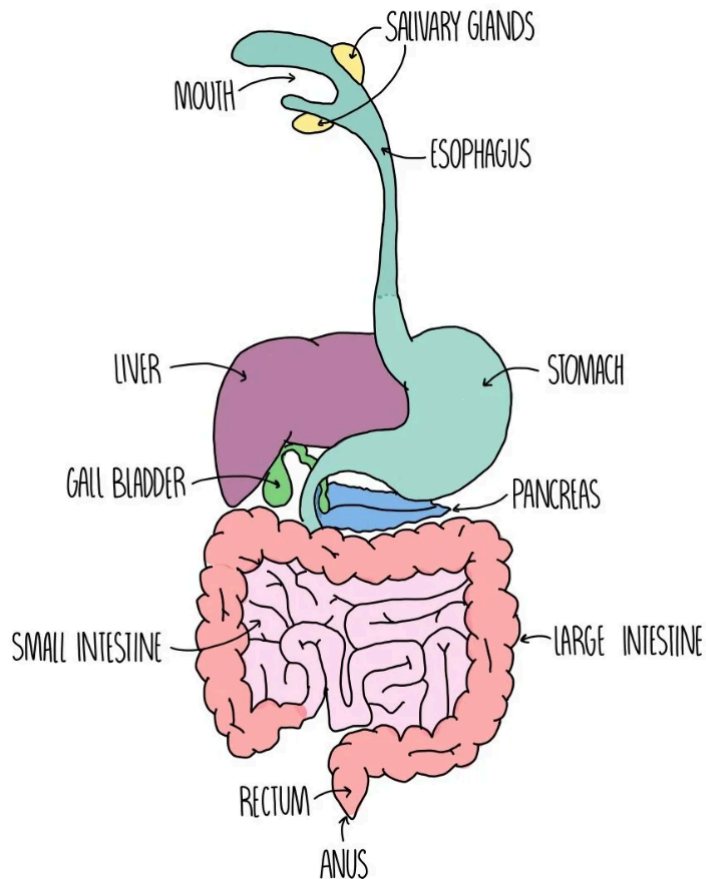
All living things are made from **cells**.

Word	Meaning
Tissue	A group of cells doing the same job (e.g. muscle tissue)
Organ	Made of different tissues (e.g. stomach)
Organ system	Group of organs working together (e.g. digestive system)

2. The Digestive System

What is Digestion?

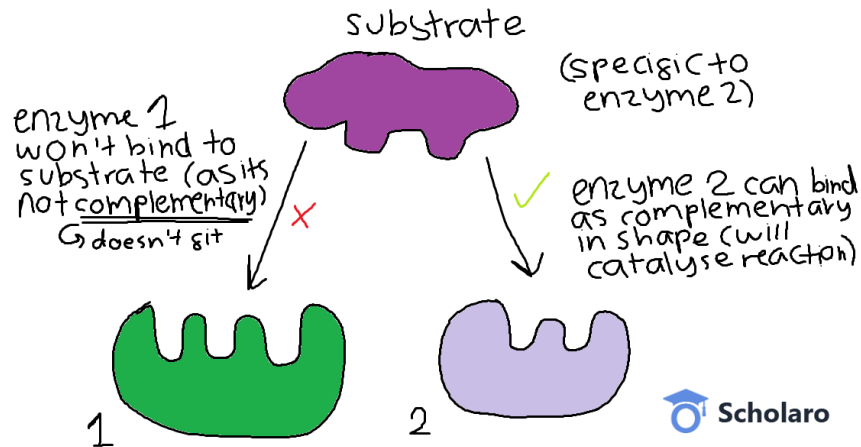
- Breaks large, **insoluble** food into small, **soluble** pieces.
- Happens in the **digestive system**.
- The “food pipe” is known as the oesophagus (not esophagus, that's the american spelling!)



Main Organs:

Organ	Job
Salivary glands	Make saliva with enzymes
Stomach	Acid kills bacteria; helps protease enzyme
Liver	Makes bile (helps break fats)
Gall bladder	Stores bile
Pancreas	Makes digestive enzymes
Small intestine	Absorbs food into blood
Large intestine	Takes water out of food
Rectum & Anus	Removes faeces

Enzymes: Special Proteins



- Help break or build molecules.
- Have a **special shape** called the **active site**.
- Like a **lock and key**.

⚠ Enzymes only work best at:

- **37°C** (body temp)
- Right **pH** (e.g. stomach enzymes like acid, others prefer neutral)

Too hot or wrong pH = **denatured** (they break and stop working)

🍞 Types of Enzymes:

Enzyme	Breaks Down	Into	Where
Amylase	Starch	Sugars	Saliva, pancreas, small intestine
Protease	Proteins	Amino acids	Stomach, pancreas, small intestine
Lipase	Fats (lipids)	Fatty acids + glycerol	Pancreas, small intestine

💡 Bile breaks fats into smaller drops = faster digestion

💡 Food Tests:

Test	Shows	Colour if Present
Benedict's	Sugar	Red/Orange

Iodine	Starch	Blue-Black
Biuret	Protein	Purple
Emulsion (ethanol)	Fat	Cloudy layer
Sudan III	Fat	Red layer on top

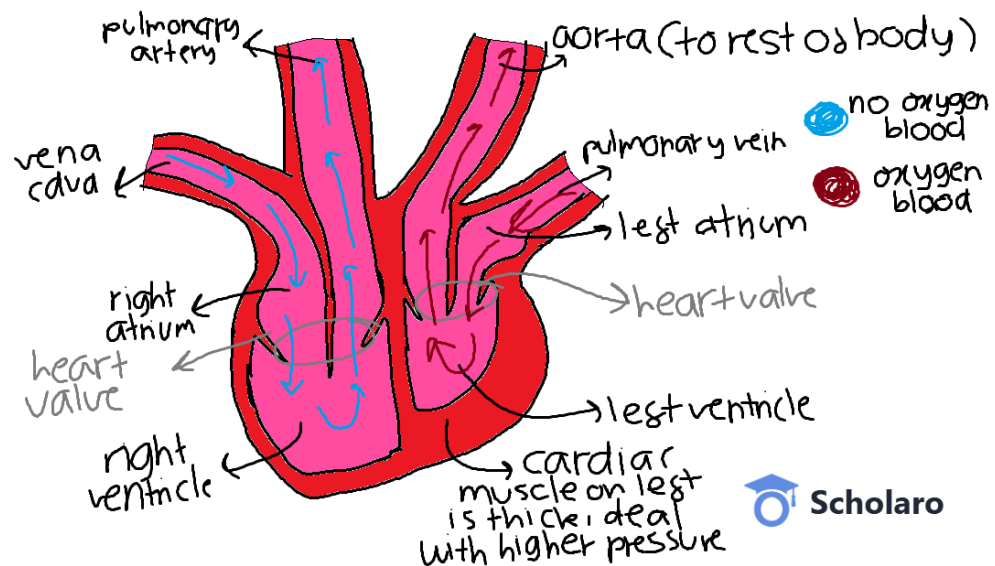
Enzyme Required Practical: Effect of pH

- Use **iodine** to check for starch
- Mix starch + enzyme + buffer (different pH)
- Test drops in spotting tile every 30s
- Stop when no more starch (iodine stays brown)
- Use: $\text{Rate} = 1000 \div \text{Time (s)}$

3. The Heart and Circulation

The Heart

- Organ that **pumps blood** in **2 circuits** (double circulation)



Side	Blood Type	Sends To
Right	Deoxygenated	To lungs

Left

Oxygenated

To body

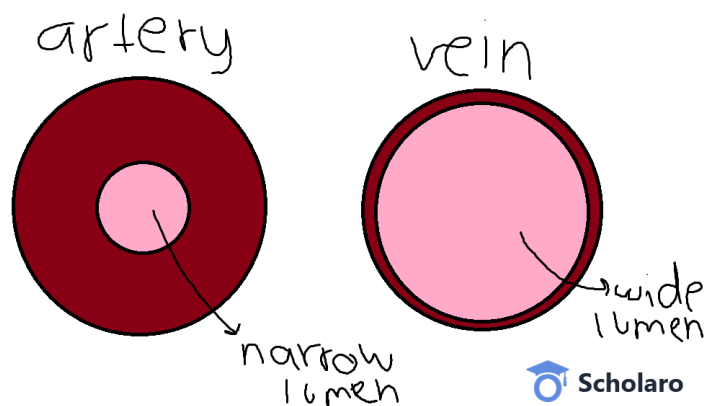
Blood flows:

1. **Vena cava** → **right atrium** → **right ventricle** → **pulmonary artery** → **lungs**
2. **Pulmonary vein** → **left atrium** → **left ventricle** → **aorta** → **body**

🧠 The **pacemaker** (in right atrium) controls heartbeat.

⚙️ If it fails, doctors use an **artificial pacemaker**.

🥁 Blood Vessels



Type	Job	Features
Arteries	Away from heart	Thick muscle, elastic
Veins	To heart	Thin walls, valves
Capillaries	Near cells	Very thin, allow diffusion

📦 Blood flow = $\text{Volume} \div \text{Time}$

👉 Lungs and Breathing

- Located in **thorax**
- Gas exchange happens in **alveoli**

Breathing (Ventilation):

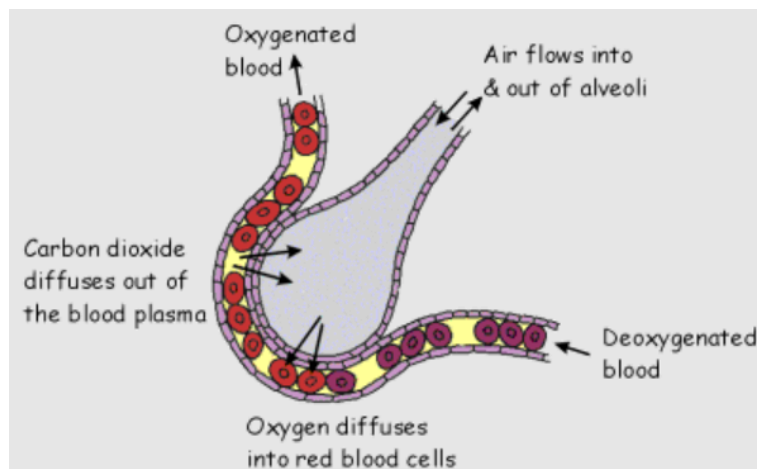
1. **Inhale:** Ribs up, diaphragm down → more space → air in
2. **Exhale:** Ribs down, diaphragm up → air out

Gas Exchange in Alveoli:

Gas	Direction	Why
Oxygen	Into blood	Low oxygen in blood
Carbon dioxide	Out to alveoli	High CO ₂ in blood

Alveoli are:

- Tiny and many → large surface area
- Close to blood
- Thin walls → fast diffusion



4. Blood and Health

What is Blood Made Of?

Part	Job
Plasma	Carries everything (glucose, CO ₂ , etc.)
Red blood cells	Carry oxygen (has haemoglobin)
White blood cells	Fight infection (antibodies, antitoxins, eat germs)
Platelets	Help blood clot, stop bleeding

5. Cardiovascular Disease

Coronary Heart Disease

- Arteries get **blocked with fat**
- Less oxygen → **heart attack**

Treatments:

Treatment	How it Helps	Problem
Stents	Open artery	Surgery risk
Statins	Lower bad cholesterol	Side effects, slow results

Valve Problems

- Blood leaks or gets stuck

Fix	Pro	Con
Biological valve	Natural	Doesn't last long
Mechanical valve	Long-lasting	Need meds (anti-clot)

Heart Failure

- Fix with **heart transplant** or **artificial heart**
- Risks: infection, blood clots, breakdown

Blood Loss

- Use **artificial blood** (temporary)
 - Buys time to get transfusion
-

6. Health & Disease

Word	Meaning
Health	Physical, mental, and social wellbeing
Disease	A problem in the body or mind

Types of Disease:

Type	Example
Communicable (infectious)	Flu, COVID

Non-communicable

Cancer, heart disease

Diseases Can Affect Each Other:

- Virus → cancer (e.g. HPV → cervical cancer)
- Infection → weak immune system → other disease
- Bad health → depression

Lifestyle and Health:

Factor	Effect
Diet	Too much/little food = diabetes, obesity
Stress	Can cause illness
Life situation	Poor hygiene or money = more disease

7. Non-Communicable Diseases

Risk Factors:

Factor	Disease
Smoking	Lung disease, cancer, heart disease
Obesity	Diabetes, heart disease
Alcohol	Liver damage, brain problems
UV light	Skin cancer
Genes	Can increase cancer risk

8. Cancer

What is Cancer?

- When cells grow **out of control**

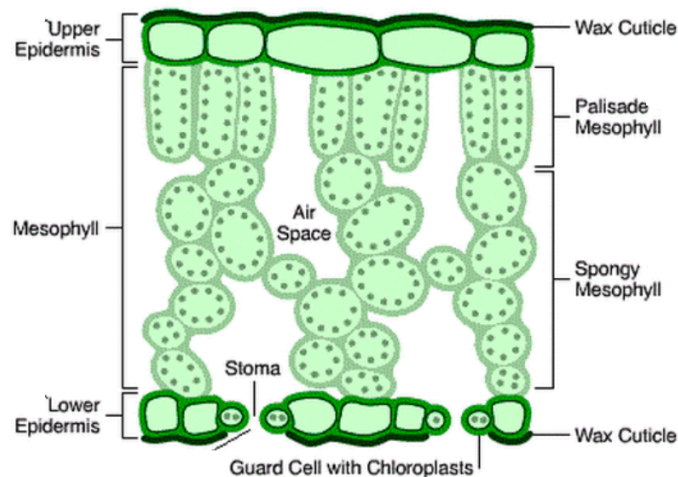
Type	Description
Benign	Stays in one place, not cancer

Malignant

Spreads, is cancer

🌿 9. Plant Tissues and Transport

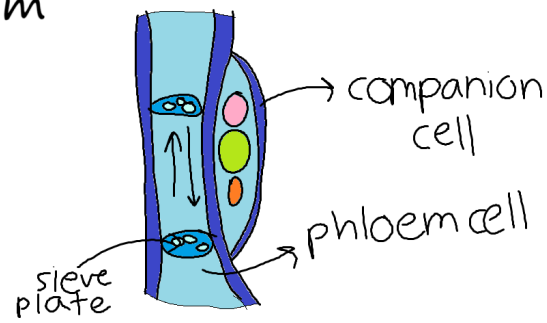
🌱 Plant Tissues:



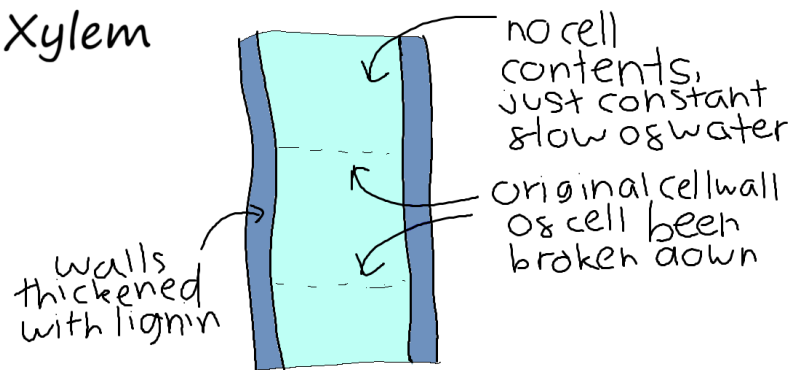
Tissue	Job
Epidermis	Covers surface, waxy, stops water loss
Palisade	Lots of chloroplasts, does photosynthesis
Spongy mesophyll	Has air spaces for gas exchange
Xylem	Carries water up, strong, has lignin
Phloem	Carries food, both directions, has sieve plates
Meristem	Can become any type of plant cell

🌳 Plant Transport Systems:

Phloem



Xylem



- **Xylem:** Moves water/minerals up (transpiration stream)
- **Phloem:** Moves food (translocation)

Transpiration:

Water lost through **stomata**.

Factor	Effect
High temp	More evaporation
More light	More stomata open
More wind	Faster diffusion
High humidity	Slower diffusion

 Use a **potometer** to measure water loss.

Summary

 You've learned:

- How cells build up into systems
- What enzymes are and how they work
- How blood and the heart support life
- What diseases are and how lifestyle affects them
- How plants transport water and food

You got this. Keep studying one step at a time!

Made with  by Scholaro for ESL students preparing for AQA Biology GCSE.